From a double well to a single barrier: the S_N2 reaction $F^- + CH_3Cl$ in the gas phase and in water

Martín Llorens Giménez*

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1 Introduction

An S_N2 reaction is a single, concerted step in which a nucleophile attacks an sp^3 carbon from the side opposite to the leaving group while the bond to it breaks, so that the configuration is inverted (Walden inversion) [1]. By default, Gaussian treats molecules as isolated, that is, in the gas phase. Under these conditions, the ion and the molecule attract each other before reacting and form ion–dipole complexes, and the reaction profile shows two wells [2]. In solution, however, the S_N2 reaction of F^- in a protic solvent such as water is known to be slow, since F^- is strongly solvated [1].

The hypothesis of this work is that water, which stabilises ions more strongly the more concentrated their charge is, will stabilise F^- much more than the transition state (TS), where the charge is shared between F and Cl. As a result, the ion–dipole complexes should disappear and the TS should rise above the separated reactants: the double well should turn into a barrier.

To test it, the potential energy surface (PES) of the reaction $F^- + CH_3Cl \longrightarrow CH_3F + Cl^-$ was computed (Exercise 3) and used to locate the TS (Exercise 4, the exercise reported here). As an extension beyond the exercise, the calculations were repeated with a continuum solvent model (PCM, water), and the energy profiles, solvation energies and rate constants in both media were compared.

2 Methods

All calculations were performed with Gaussian 09 [3] at the HF/6-31+G* level. The diffuse functions (+) are required to describe the extended electron density of the anions, and the polarisation functions (*) add d functions on the heavy atoms. All species are closed-shell singlets. Reactants, products and the two ion–dipole complexes were optimized and characterised by harmonic frequency calculations.

The PES was obtained through a relaxed scan of d(C–F) from 1.3 to 2.9 Å and d(C–Cl) from 3.6 to 1.6 Å, in steps of 0.2 Å (99 points), optimizing all other coordinates at each point; a dummy atom was used to define the collinear F–C–Cl arrangement without 180° angles. The TS search started from the saddle point of the scan (2.2 Å, 2.1 Å) with a planar CH_3 group (H–C–F = 90°), using `opt=(ts,calcfc,noeigentest)`, and was confirmed by a single imaginary frequency. Solvent effects were modelled with the polarizable continuum model [4] (`scrf=(pcm,solvent=water)`), re-optimizing all species and repeating the relaxed scan. Energies are given at 298.15 K and 1 atm (1 E_h = 627.51 kcal/mol). Rate constants were obtained from transition state theory, $k = (\frac{k_B T}{h}) \cdot (\frac{RT}{P^\circ})^{n-1} \cdot e^{(-\Delta G^\ddagger / RT)}$, where n is the molecularity of the step.

3 Results and discussion

Gas phase: a double well. The gas-phase PES (Figure 1a) shows two closed valleys, which coincide with the independently optimized ion–dipole complexes (RC and PC), and a saddle point between them. The scan energies reproduce the reference data provided with the session, which validates the procedure. The optimized TS has a single imaginary frequency ($442i$ cm^{-1}), whose normal mode is the S_N2 itself: F approaches C, Cl leaves and the CH_3 umbrella inverts. Its geometry is collinear and almost symmetric (d(C–F) = 2.125 Å, d(C–Cl) = 2.133 Å), and matches the saddle point of the scan to 0.01 Å. However, the H–C–F angle (82.7°) is closer to the reactant complex (∼72°) than to the product complex (∼109°): as expected from Hammond's postulate for a strongly exergonic reaction, the TS is early and reactant-like.

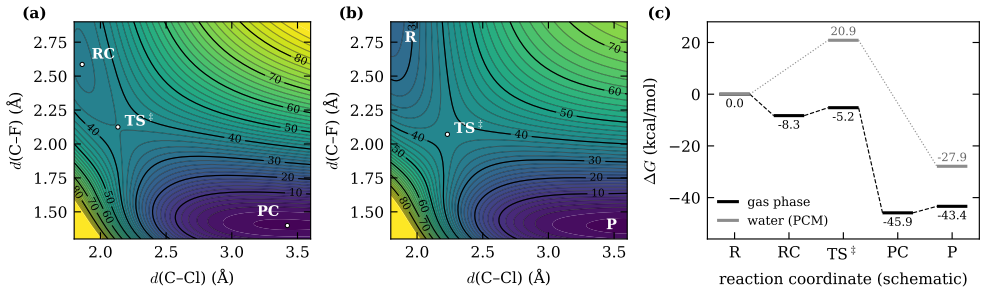


Figure 1: PES of the reaction at the HF/6-31+G* level in (a) the gas phase and (b) water (PCM), from relaxed scans (energies relative to the lowest point of each scan; contours every 2 kcal/mol; d(C–Cl) ≥ 1.8 Å). RC, PC and TS‡ mark the optimized reactant complex, product complex and transition state; R and P, the reactant and product valleys. (c) Free-energy profiles relative to the separated reactants (R); RC and PC are not minima in water.

*__Declaration of AI use:__ An AI assistant (Claude Opus 5.5, Anthropic) was used for guidance to help structure the text, which the author then wrote independently. AI was mainly used to revise English expressions and grammar, and to write the Python (matplotlib) code for figures and tables. All calculations, data extraction, and final decisions on the content are the author's own.

The scan, however, only provides the shape of the surface. With a grid step of 0.2 Å, it does not contain the exact stationary points: the lowest point of the product valley, at (3.4, 1.5 Å), lies 3.1 kcal/mol above the optimized product complex, whose C–F distance (1.40 Å) falls between two grid points. All the energies discussed below therefore come from full optimizations of the minima and the TS, not from the scan.

Table 1 and Figure 1c show the resulting profile. The reactant complex lies 14.5 kcal/mol below the separated reactants (ΔE), and the TS only 3.0 kcal/mol above the complex, that is, 11.5 kcal/mol *below* the separated reactants: there is no net barrier, in agreement with the double-well model [2]. In free energy, the TS rises to -5.2 kcal/mol, because joining two particles into one lowers the entropy ($-T\Delta S \approx +6$ kcal/mol). From the reactant complex ($\Delta G^{\ddagger} = 3.06$ kcal/mol, $n = 1$), $k = 3.5 \times 10^{10}$ s^{-1}.

Table 1: Energies relative to $F^- + CH_3Cl$ (kcal/mol) in the gas phase and in water (PCM).

	Gas phase			Water (PCM)	
Species	ΔE	ΔH	ΔG	ΔE	ΔG
$F^- + CH_3Cl$	0.00	0.00	0.00	0.00	0.00
$F{-}{\cdots}CH_3Cl$	−14.51	−14.39	−8.29	–	–
TS	−11.52	−12.03	−5.22	14.03	20.86
$FCH_3{\cdots}Cl^-$	−52.68	−51.43	−45.89	–	–
$CH_3F + Cl^-$	−44.64	−43.64	−43.37	−29.13	−27.88

Water: a single barrier. With PCM, neither complex survives: on re-optimization, the ions move away to 4.5–5.1 Å, where the surface is flat, because the dielectric screens the ion–dipole attraction. In the water PES (Figure 1b), the closed wells of the gas phase become open valleys: the energy keeps decreasing as either ion moves away. The TS, instead, remains a first-order saddle point ($547i$ cm^{-1}, higher than in the gas phase: the barrier is not only higher but also narrower) and now lies *above* the reactants: $\Delta G^{\ddagger} = 20.9$ kcal/mol, which gives $k = 7.8 \times 10^{-2}$ M^{-1} s^{-1} ($n = 2$). The double well has become a single barrier, as predicted. The solvation energies, $E(\text{water}) - E(\text{gas})$, explain why: -86.2 kcal/mol for F^-, -70.5 for Cl^- and -63.2 for the TS (only -2.5 for CH_3Cl). The smaller the ion, the larger its solvation, as in the Born model ($\propto -q^2/r$); in the TS, the charge is shared between F and Cl. The reactants are therefore stabilised by 25.6 kcal/mol more than the TS, which turns -11.5 into $+14.0$ kcal/mol. For the same reason (F^- is better solvated than Cl^-), the reaction is less exergonic in water, and the TS moves towards the products (d(C–F) = 2.071 Å, d(C–Cl) = 2.231 Å, H–C–F = 86.0°), again as Hammond's postulate predicts.

These results are qualitative. PCM only accounts for the dielectric constant of water and does not include hydrogen bonds, which are especially strong for F^-; the real solvation of F^- and the real barrier should therefore be even larger. In addition, HF neglects electron correlation.

4 Conclusion

1. In the gas phase, the S_N2 reaction $F^- + CH_3Cl$ follows a double-well profile in which the TS ($442i$ cm^{-1}) lies 11.5 kcal/mol *below* the separated reactants: there is no net barrier, and the only barrier, 3.1 kcal/mol in ΔG, is the one from the reactant ion–dipole complex.
2. Water (PCM) changes the profile completely: the ion–dipole complexes disappear and a barrier of 20.9 kcal/mol appears. This is because the solvent stabilises F^- (-86.2 kcal/mol), whose charge is concentrated, much more than the TS (-63.2 kcal/mol), where it is shared between F and Cl, as hypothesised. Hydrogen bonds, absent in PCM, would make the barrier even higher.
3. As a consequence, the mechanism and the rate change: in the gas phase, the crossing of the TS is a unimolecular step from the complex, with $k = 3.5 \times 10^{10}$ s^{-1}, whereas in water it is a bimolecular reaction between the separated reactants, with $k = 7.8 \times 10^{-2}$ M^{-1} s^{-1}, which is slow at room temperature.
4. The TS geometry is consistent with Hammond's postulate in both media: it is early and reactant-like in the gas phase (H–C–F = 82.7°), and moves towards the products in water (86.0°), where the reaction is less exergonic.

References

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